

Experimental VLE, consistency and fitting

After midterm · Session 2 of 6 · 180 minutes

Soorathep Kheawhom

30 September 2026

Retrieve and check · First 12 minutes

At 350 K and $z_1=0.5$, use $P_{sat1}=150$ and $P_{sat2}=60$ kPa.

- **3 min, alone:** predict the phase at 80, 95 and 110 kPa.
- **5 min:** at 95 kPa, check $z_1=(1-\beta)x_1+\beta y_1$ using $x_1=0.388889$, $y_1=0.614035$, $\beta=0.493506$. Check both composition sums.
- **2 min:** each person opens [Lab 01](#) and saves a baseline.
- **2 min:** compare checks and summarize in Thai.

แยกองค์ประกอบ x , y , z ออกจากสัดส่วนไธ β แล้วตรวจสอบสมดุลมวล

Before class: 3-page reading · 15–20 minutes

Bring the [Week 2 worksheet](#). Today: does a small fitting error establish trust?

The 180-minute class

ACTIVITY	MINUTES
Opening question and essential concepts	30
One hand calculation	20
One lab demonstration	20
Break	10
Student pair investigation	70
Discussion, exit ticket and feedback	30

The opening 30 minutes include the 12-minute retrieval activity. Pair work stays 70 minutes. Final block: discussion 20, exit ticket 5, feedback 3, closing 2 minutes.

Activity and its reference

$$\mu_i = \mu_i^\circ + RT \ln a_i, \quad a_i = \gamma_i x_i$$

The activity is dimensionless; γ is its correction relative to an ideal liquid solution using the pure-liquid standard.

For $x_1=0.4$ and $\gamma_1=1.25$, $a_1=0.50$. Residual and excess reference comparisons are in the reading and appendix.

activity ไม่มีหน่วย และ a ไม่ใช่ γ ต้องคูณด้วยองค์ประกอบ x

Properties → prediction → fitting

1. Verify the Antoine form, units and applicable temperature range.
2. Predict P and y from a chosen liquid model.
3. Infer activities from observations, then compare model fits.

$$\gamma_i = \frac{y_i P}{x_i P_i^{sat}(T)}$$

This inversion assumes ideal vapor and negligible Poynting correction. Convergence, consistency and prediction are separate checks.

ตรวจสอบสมการ Antoine หน่วย และช่วงอุณหภูมิ ก่อนเชื่อค่าที่ fit ได้

A vapor-pressure error becomes an activity error

Synthetic point: $P=100$ kPa, $x_1=0.4$, $y_1=0.6$ and $P_1^{\text{sat}}=120$ kPa.

$$\gamma_1 = \frac{0.6(100)}{0.4(120)} = 1.25$$

Using 150 kPa instead yields $\gamma_1=1.00$. The measured mixture data did not change; the inferred liquid behavior did.

Pause: calculate a_1 for each case before comparing answers.

ข้อมูลสารผสมเดิม แต่ใส่ความดันไอผิด ก็ทำให้แปลผล γ ผิดได้

An independent check

Work alone first, then exchange checks using the worksheet.

1. Write the Antoine equation, temperature unit and pressure unit; evaluate one P_{sat} .
2. Substitute one observed T , P , x , y point into $\gamma = yP/(xP_{\text{sat}})$; show units cancelling.
3. Check $x_1 + x_2 = 1$ and $y_1 + y_2 = 1$. Compare the same observations and property convention.

เขียนแทนค่าจริงหนึ่งจุด การบอกว่า “ได้ตรงกับ lab” ยังไม่แสดงวิธีตรวจ

One demonstration • 20 minutes

Open the core lab

Load and validate the synthetic example in Lab 08. Inspect its Antoine form and units, then fit one-parameter Margules. Compare predicted P and y with one observed point, then inspect the residual pattern.

Each person repeats the baseline and saves it. Pause here to resolve setup problems before model comparison.

Pair investigation • 70 minutes

- 10 min: each person loads and validates the same baseline.
- 25 min: change one input and calculate.
- 20 min: calculate γ individually, then exchange checks and roles.
- 15 min: prepare one plot or table and a short explanation.

In Lab 08, validate the supplied synthetic example and compare one-parameter Margules with NRTL at $\alpha=0.3$ on the same observations. Report one residual pattern and verify one activity coefficient by hand.

Discussion • 20 minutes

Does a small fitting error mean the data and model are trustworthy?

Compare one residual pattern and one numerical check. Distinguish a converged fit, a consistency diagnostic and prediction on excluded data.

fit ได้ดี ไม่ได้ยืนยันว่าข้อมูลถูกหรือทำนายนอกข้อมูลได้ดี ต้องตรวจแยกกัน

After class • 30–40 minutes

In Lab 08, validate the supplied synthetic example and compare one-parameter Margules with NRTL at $\alpha=0.3$ on the same observations. Report one residual pattern and verify one activity coefficient by hand.

Assessment 2 and independent checkpoint

Retain one study JSON, one comparison and the completed [worksheet](#). Finish the in-class work; no additional dataset is required.

Optional: [Session appendix](#).

[Learning path](#) • [Offline package](#)

Exit ticket + feedback • 8 minutes

5 min • Explain and check

Answer the session question. In the evidence box, show one substituted calculation of γ or P_{sat} with units, not just the final number.

3 min • Help improve the next class

Rate the pace, report lab access, and suggest one improvement.

Thai or English. No name, ID or email requested.

Wait for “**Response received.**” We will use the final 2 minutes to close the class.



Scan to submit Session 2 exit ticket and feedback.

[Open Session 2 form](#)